Supplementary Information: Genome-Scale Context, Stoichiometric Refinement, and Neuro-Symbolic Reasoning for Proteome Function Annotation

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S1 Hyperparameter specification and one-at-a-time sensitivity analysis

All 28 primary pipeline constants across Stage 1 (iterative Bayesian log-odds refiner, alongside fixed convergence controls $K_{\max} = 6$ and $\varepsilon_{\text{iter}} = 0.005$ in `IterativeRefiner.groovy`), Stage 2A (deterministic symbolic enforcer, alongside candidate partner gate 0.08 and singleton prune value 0.45 in `ConsistencyEnforcer.groovy`), and Stage 2B (stoichiometric locus refiner, alongside fixed elemental-imbalance penalty $w_{\text{imb}} = 100.0$) were fitted exclusively on the *Escherichia coli* K-12 MG1655 development proteome via two-stage block coordinate search and held frozen across all out-of-sample prokaryotic, archaeal, viral, and eukaryotic benchmarks. Supplementary Table S1 reports the complete parameter inventory, default values, and one-at-a-time ±20% perturbation sensitivity across the 8 out-of-sample reference genomes of Table 1, the 144 held-out metabolic reactions in *Bacillus subtilis* 168 (76 covered by candidate pools), and the 22-genome `Price-149` RB-TnSeq enzyme benchmark (source: `results/hyperparameter_sensitivity_non_ecoli.tsv`). Across all 28 parameters, a ±20% perturbation changes the 8-genome out-of-sample micro $F_{\max}$ by at most 0.0009, the internal GAEF composite score by at most 0.0038, *B. subtilis* LRO Hit@1 by at most 1.32 percentage points, and `Price-149` CLEAN multi-predictor Iter 3 $F_{\max}$ (0.6084; single-predictor Iter 3 Top-1 = 63/148, $F_{\max} = 0.5214$, dataset macro $F_{\max} = 0.4216$) by at most 0.0042.

S2 Cross-ontology taxonomic and anatomical constraint bridge

Supplementary Table S2 summarizes the five resource tables programmatically extracted from the official OBO Foundry releases of the Gene Ontology (GO), Uberon multi-species anatomy ontology, Plant Ontology (PO), Cell Ontology (CL), and NCBI Taxonomy using the OWL API and ELK reasoner (`gspa-core/src/main/resources/gspa/`).

Table S1. Complete inventory of the 28 GSPA v2.1.0 hyperparameters and one-at-a-time ±20% sensitivity analysis on out-of-sample non-*E. coli* benchmarks (source: `results/hyperparameter_sensitivity_non_ecoli.tsv`).

Stage / Module	Symbol	Parameter Name	Role in Inference	Default	Primary Metric	−20% Metric	Default Metric	+20% Metric	Max $\|\Delta\|$
Stage 1 (B_1)	α_1	`essentiality_weight`	Essentiality log-odds prior scale	1.50	8-genome micro $F_{\max}$	0.7747	0.7749	0.7750	0.0002
Stage 1 (B_1)	θ_{ms}	`essentiality_threshold`	Unfulfilled essential function gate	0.50	8-genome micro $F_{\max}$	0.7746	0.7749	0.7751	0.0003
Stage 1 (Iter)	γ_{iter}	`iterative_damping`	Iterative log-odds update damping factor	0.50	8-genome micro $F_{\max}$	0.7748	0.7749	0.7748	0.0001
Stage 1 (B_2)	α_2	`coherence_weight`	Pathway and complex coherence log-odds scale	1.00	8-genome micro $F_{\max}$	0.7744	0.7749	0.7753	0.0005
Stage 1 (B_2)	θ_{coh}	`coherence_coverage_gate`	Partner coverage threshold for coherence boost	0.70	8-genome micro $F_{\max}$	0.7741	0.7749	0.7752	0.0008
Stage 1 (B_2)	β_{path}	`pathway_boost_scale`	Pathway completion deficit scale	1.50	8-genome micro $F_{\max}$	0.7746	0.7749	0.7751	0.0003
Stage 1 (B_2)	β_{proc}	`process_boost_scale`	Biological process completion deficit scale	1.00	8-genome micro $F_{\max}$	0.7747	0.7749	0.7750	0.0002
Stage 1 (B_2)	β_{single}	`singleton_penalty`	Obligate heteromeric singleton log-odds penalty	0.75	8-genome micro $F_{\max}$	0.7747	0.7749	0.7751	0.0002
Stage 1 (B_2)	θ_{het}	`heteromeric_min_coverage`	Minimum subunit coverage before singleton penalty	0.35	8-genome micro $F_{\max}$	0.7746	0.7749	0.7750	0.0003
Stage 1 (B_3)	α_3	`consistency_weight`	SAT4J MUS taxonomic violation log-odds penalty	3.00	8-genome micro $F_{\max}$	0.7748	0.7749	0.7749	0.0001
Stage 1 (B_4)	α_4	`anatomical_penalty`	Forbidden Uberon/PO/CL structure log-odds penalty	3.50	8-genome micro $F_{\max}$	0.7749	0.7749	0.7749	0.0000
Stage 1 (B_4)	α_{boost}	`anatomical_dev_boost`	Active-structure developmental rescue log-odds boost	0.65	6-eukaryote CAFA $F_{\max}$	0.7112	0.7117	0.7119	0.0005
Stage 1 (B_4)	θ_{dev}	`anatomical_dev_support_gate`	Minimum supporting proteins for developmental rescue	0.50	6-eukaryote CAFA $F_{\max}$	0.7114	0.7117	0.7115	0.0003
Stage 1 (B_5)	α_5	`gap_filling_weight`	Metabolic gap-filling RLGC log-odds scale	1.50	*B. subtilis* LRO Hit@1	0.4236	0.4306	0.4306	0.0070
Stage 1 (B_5)	λ_{BF}	`rlgc_lambda_bf`	Operon empirical Bayes factor weight in RLGC	0.35	*B. subtilis* LRO Hit@1	0.4236	0.4306	0.4375	0.0069
Stage 1 (B_5)	λ_{dom}	`rlgc_lambda_dom`	Pfam domain EC compatibility weight in RLGC	0.20	*B. subtilis* LRO Hit@1	0.4306	0.4306	0.4306	0.0000
Stage 1 (B_5)	λ_{PLM}	`rlgc_lambda_plm`	ESM-2 reaction-centroid cosine weight in RLGC	0.45	*B. subtilis* LRO Hit@1	0.4236	0.4306	0.4375	0.0069
Stage 1 (B_6)	α_6	`genomic_context_weight`	Co-directional operon partner log-odds scale	1.00	8-genome micro $F_{\max}$	0.7745	0.7749	0.7752	0.0004
Stage 1 (B_6)	$d_{\max}$	`operon_max_distance_bp`	Maximum same-strand intergenic distance (bp)	150	8-genome micro $F_{\max}$	0.7746	0.7749	0.7751	0.0003
Stage 2A	γ_{relax}	`relax_retention_factor`	Probability retention factor on valid GO ancestor	1.00	6-eukaryote CAFA $F_{\max}$	0.7105	0.7118	0.7118	0.0013
Stage 2B	θ_{sub}	`subunit_bottleneck_gate`	Subunit conjunction geometric-mean bottleneck gate	0.25	`Price-149` CLEAN $F_{\max}$	0.6072	0.6084	0.6080	0.0012
Stage 2B	θ_{core}	`core_reaction_threshold`	Draft core reaction inclusion threshold $S_{\text{rxn}}(j)$	0.50	`Price-149` CLEAN $F_{\max}$	0.6054	0.6084	0.6068	0.0030
Stage 2B	α_{bio}	`biomass_fraction_floor`	Minimum required fraction of $v_{\text{bio}}^{\max}$ in pFBA	0.10	`Price-149` CLEAN $F_{\max}$	0.6082	0.6084	0.6083	0.0002
Stage 2B	η_{null}	`null_hypothesis_mass`	Unassigned-gene hypothesis $h_\varnothing$ prior mass	0.05	`Price-149` CLEAN $F_{\max}$	0.6075	0.6084	0.6079	0.0009
Stage 2B	γ_{FBA}	`fba_locus_boost_scale`	Stoichiometric locus responsibility log-odds scale	2.20	`Price-149` CLEAN $F_{\max}$	0.6048	0.6084	0.6095	0.0036
Stage 2B	γ_{damp}	`outer_loop_damping`	Geometric damping factor across outer passes	0.60	`Price-149` CLEAN $F_{\max}$	0.6076	0.6084	0.6088	0.0008
Stage 2B	β_1, β_2	`resp_exponents_seq_bf`	Sequence odds and operon BF exponents in $w_{j,u}(p)$	1.00	`Price-149` CLEAN $F_{\max}$	0.6050	0.6084	0.6092	0.0034
Stage 2B	β_3, β_4	`resp_weights_syn_plm`	Synteny and ESM-2 exponents in $w_{j,u}(p)$	1.50	`Price-149` CLEAN $F_{\max}$	0.6042	0.6084	0.6098	0.0042

Table S2. Summary of programmatic cross-ontology constraint tables in GSPA v2.1.0 (`gspa-core/src/main/resources/gspa/`).

Resource File	Ontologies Bridged	Data Rows	Distinct Classes	Constraint Semantics & Examples
`go-taxon-constraints.tsv`	GO ↔ NCBITaxon	26,911	18,159 GO terms	13,819 `only_in_taxon` + 13,092 `never_in_taxon` rules (e.g., `GO:0009579` thylakoid)
`go-anatomical-constraints.tsv`	GO ↔ Uberon / PO / CL	2,715	2,674 GO; 1,294 Anat.	30 object properties (`results_in_development_of`, `occurs_in`, etc.)
`uberon-po-taxon-constraints.tsv`	Uberon / PO / CL ↔ NCBITaxon	8,331	1,247 Anat. classes	4,767 `only_in_taxon` + 3,564 `never_in_taxon` rules (e.g., `UBERON:0000948` heart → Metazoa)
`uberon-po-hierarchy.tsv`	Uberon / PO / CL DAG	7,722	2,855 Anat. classes	Transitive `is_a` and `part_of` edges across anatomical structures
`ncbi-taxon-hierarchy.tsv`	NCBI Taxonomy DAG	1,818	1,305 Taxon clades	Transitive taxonomic `is_a` lineage closure for benchmark clades

S3 Extended Suite 1 and Suite 2 benchmarks

Supplementary Table S3 reports the complete 16-row post-cutoff Swiss-Prot enzyme benchmark (`Holdout-125`, $n = 125$ proteins across 293 ground-truth protein–EC pairs; source: `results/scenario1_swissprot_ec_holdout125_real.tsv`), separating standalone per-protein baselines, single-predictor refinement (`DeepProZyme + METEOR v8` and `DeepProZyme + GSPA Single-Pred`), and 3-method ensemble pooling (`Noisy-OR` and `GSPA Iter 1-3`). Because these 125 proteins are dispersed across 74 distinct species and evaluated as isolated sequences, single-predictor and ensemble refinement raise Top-3 recall to 43.2% (54/125) and Top-5 recall to 44.0% (55/125) while preserving Top-1 accuracy (32.0%–33.6% vs. 32.8% for standalone CLEAN). Supplementary Table S4 reports the complete 12-row enzyme evaluation across the 95-genome bacterial panel under four donor-database homology-masking tiers (`vanilla`, $\leq 70\%$, $\leq 50\%$, and $\leq 30\%$ sequence identity; source: `results/scenario1_95genomes_ec_homology_real.tsv`), explicitly separating single-predictor `+ GSPA Single-Pred` from 3-method `+ GSPA Multi-Pred` alongside exact discordant protein counts (b_{01}, b_{10}) and two-sided exact McNemar p-values.

Table S3. Complete post-cutoff Swiss-Prot enzyme evaluation (`Holdout-125`, $n = 125$ proteins, 293 ground-truth EC annotations; source: `results/scenario1_swissprot_ec_holdout125_real.tsv`).

Predictor / Refinement Method	Masking Tier	Top-1 Acc. (Hits)	Top-3 Acc. (Hits)	Top-5 Acc. (Hits)	MRR	Micro $F_{\max}$
CLEAN (2023)	`vanilla`	32.8% (41/125)	40.8% (51/125)	41.6% (52/125)	0.3719	0.4360
DeepProZyme (2023)	`vanilla`	23.2% (29/125)	31.2% (39/125)	32.0% (40/125)	0.2750	0.2715
DeepProZyme + METEOR v8 (Single-Pred)	`vanilla`	25.6% (32/125)	31.2% (39/125)	31.2% (39/125)	0.2872	0.2804
DeepProZyme + GSPA Single-Pred (SLR + Coh.)	`vanilla`	23.2% (29/125)	31.2% (39/125)	32.8% (41/125)	0.2779	0.2703
EnzBERT (2023)	`vanilla`	32.8% (41/125)	37.6% (47/125)	39.2% (49/125)	0.3531	0.3596
Noisy-OR (3-Method Ensemble)	`vanilla`	33.6% (42/125)	40.8% (51/125)	43.2% (54/125)	0.3807	0.4381
GSPA (3-Method, Iter 1)	`vanilla`	32.0% (40/125)	42.4% (53/125)	44.0% (55/125)	0.3775	0.4035
GSPA (3-Method, Iter 2)	`vanilla`	32.0% (40/125)	42.4% (53/125)	44.0% (55/125)	0.3775	0.3891
GSPA (3-Method, Iter 3)	`vanilla`	32.0% (40/125)	43.2% (54/125)	44.0% (55/125)	0.3781	0.3830
EnzBERT	`filt70` ($\leq 70\%$)	30.4% (38/125)	35.2% (44/125)	35.2% (44/125)	0.3305	0.3204
EnzBERT	`filt50` ($\leq 50\%$)	28.0% (35/125)	33.6% (42/125)	33.6% (42/125)	0.3040	0.3163
EnzBERT	`filt30` ($\leq 30\%$)	17.6% (22/125)	24.8% (31/125)	24.8% (31/125)	0.2139	0.2212
CLEAN	`filt30` ($\leq 30\%$)	24.0% (30/125)	30.4% (38/125)	32.8% (41/125)	0.2741	0.2299
DeepProZyme	`filt70` ($\leq 70\%$)	23.2% (29/125)	31.2% (39/125)	32.0% (40/125)	0.2713	0.2715
DeepProZyme	`filt50` ($\leq 50\%$)	23.2% (29/125)	31.2% (39/125)	32.8% (41/125)	0.2720	0.2684
DeepProZyme	`filt30` ($\leq 30\%$)	22.4% (28/125)	31.2% (39/125)	32.8% (41/125)	0.2674	0.2650

Table S4. Single-predictor and multi-predictor EC refinement across 95 bacterial genomes under four donor-database homology-masking tiers (source: `results/scenario1_95genomes_ec_homology_real.tsv`). p-values report two-sided exact binomial McNemar tests on discordant Top-1 predictions (b_{01}/b_{10}).

Predictor	Tier	Genomes / Proteins	Base Top-1 ($F_{\max}$)	METEOR v8 Top-1 (Δ)	METEOR b_{01}/b_{10} (p)	GSPA Single-Pred Top-1 (Δ)	Single-Pred b_{01}/b_{10} (p)	GSPA Multi-Pred Top-1 (Δ)	Multi-Pred b_{01}/b_{10} ($F_{\max}$)
CLEAN	vanilla	73/52,814	0.8214 (0.8796)	0.8234 (+108)	184/76 ($< 10^{-6}$)	0.8228 (+77)	99/22 ($< 10^{-6}$)	0.8243 (+155)	162/7 (0.8816)
CLEAN	≤ 70%	95/69,663	0.8267 (0.8836)	0.8289 (+149)	191/42 ($< 10^{-6}$)	0.8283 (+108)	131/23 ($< 10^{-6}$)	0.8289 (+149)	154/5 (0.8851)
CLEAN	≤ 50%	95/69,663	0.8271 (0.8844)	0.8294 (+162)	208/46 ($< 10^{-6}$)	0.8291 (+139)	156/17 ($< 10^{-6}$)	0.8293 (+155)	158/3 (0.8859)
CLEAN	≤ 30%	94/68,681	0.8288 (0.8833)	0.8306 (+122)	156/34 ($< 10^{-6}$)	0.8302 (+93)	107/14 ($< 10^{-6}$)	0.8306 (+126)	132/6 (0.8845)
DeepProZyme	vanilla	95/69,666	0.9022 (0.9331)	0.9010 (−85)	103/188 (1.0×10^{-6})	0.9017 (−33)	24/57 (3.2×10^{-4})	0.9027 (+36)	51/15 (0.9335)
DeepProZyme	≤ 70%	95/69,666	0.8965 (0.9290)	0.8961 (−27)	151/178 (0.152)	0.8963 (−15)	44/59 (0.167)	0.8975 (+67)	78/11 (0.9296)
DeepProZyme	≤ 50%	95/69,666	0.8966 (0.9288)	0.8966 (+1)	163/162 (1.000)	0.8966 (−3)	49/52 (0.842)	0.8979 (+87)	93/6 (0.9297)
DeepProZyme	≤ 30%	95/69,666	0.8967 (0.9289)	0.8970 (+19)	170/151 (0.315)	0.8966 (−4)	51/55 (0.771)	0.8976 (+67)	76/9 (0.9296)
EnzBERT	vanilla	95/69,666	0.8430 (0.8707)	0.8416 (−92)	100/192 ($< 10^{-6}$)	0.8428 (−10)	51/61 (0.395)	0.8438 (+59)	65/6 (0.8713)
EnzBERT	≤ 70%	95/69,666	0.8394 (0.8675)	0.8374 (−146)	96/242 ($< 10^{-6}$)	0.8391 (−26)	41/67 (0.0157)	0.8405 (+70)	76/6 (0.8682)
EnzBERT	≤ 50%	95/69,666	0.8330 (0.8602)	0.8304 (−181)	132/313 ($< 10^{-6}$)	0.8324 (−38)	31/69 (1.8×10^{-4})	0.8343 (+89)	93/4 (0.8611)
EnzBERT	≤ 30%	95/69,666	0.8256 (0.8515)	0.8232 (−167)	108/275 ($< 10^{-6}$)	0.8254 (−18)	39/57 (0.082)	0.8267 (+78)	80/2 (0.8524)

S4 Extended stoichiometric audit, curated BiGG reference GEMs, and BacDive cross-validation

Supplementary Table S5 reports the 108-genome stoichiometric invariant audit (`results/pillar2_invariant_audit_summary.ts` the evaluation across six manually curated BiGG reference genome-scale metabolic models (`iML1515` *E. coli*, `STM_v1_0` *S. enterica*, `iYL1228` *K. pneumoniae*, `iJN1463` *P. putida*, `iYS854` *S. aureus*, and `iYO844` *B. subtilis*; source: `results/pillar2_curated6_gems_real.tsv`), and the role-stratified `Price-149` breakdown on CLEAN across Stratum A ($n = 91$), Stratum B ($n = 55$), and Stratum C ($n = 2$; source: `results/pillar2_price149_by_stratum.tsv`). Supplementary Table S6 reports the per-fold 5-fold genus-blocked cross-validation metrics in `BacDive` (source: `results/pillar3_bacdive_per_fold_cv.tsv`).

Table S5. 108-genome stoichiometric invariant audit (`results/pillar2_invariant_audit_summary.tsv`), curated BiGG reference GEM comparison ($n = 6$ species; `results/pillar2_curated6_gems_real.tsv`), and role-stratified `Price-149` CLEAN refinement across Strata A, B, and C (`results/pillar2_price149_by_stratum.tsv`).

Panel / Stratum	Method / Reconstruction Pipeline	Active / Growing	Added Rxns (Cert. / Pad.)	Imbal. / Dead-End Added	EC Prec. / Rec. / F_1	BiGG Rxn F_1	Top-1 Hits (Acc.)
108-Genome Audit	Raw Sequence Threshold (≥ 0.50)	2,809.5 (0% grow)	0.0 (0.0/0.0)	2,621.7 active / 9.75	n/a	n/a	n/a
	METEOR v8 (Negative-Cost MILP)	3,121.0 (100% grow)	313.35 (98.33/215.02)	2,597.8 active / 2.36	n/a	n/a	n/a
	Grow-Support Threshold Repair (Pos.-Cost LP)	2,584.8 (100% grow)	81.31 (81.31/0.00)	0.0 added / 2.23	n/a	n/a	n/a
	GSPA StoichiometricLocusRefiner (SLR)	2,815.2 (300.6 **flux**, 100% **grow**)	5.70 (5.70/0.00)	0.0 **added** / 0.00	n/a	n/a	n/a
6 Curated BiGG GEMs	CarveMe v1.5.2 (Native BiGG Database)	6/6 (100%)	n/a	n/a	0.8217/0.8927/0.8536	0.6955	n/a
	Reconstructor (ModelSEED Namespace)	4/6 (67%)	n/a	n/a	0.4387/0.8787/0.5844	n/a	n/a
	DeepProZyme + METEOR v8 (Neg.-Cost MILP)	6/6 (100%)	n/a	n/a	0.4383/0.8842/0.5851	0.2091	n/a
	GSPA SLR (DeepProZyme Single-Pred)	6/6 (100%)	n/a	n/a	0.4547/0.7667/0.5700	n/a (Stoich. Probe)	n/a
	GSPA SLR (CLEAN Single-Pred)	6/6 (100%)	n/a	n/a	0.4750/0.7262/0.5738	n/a (Stoich. Probe)	n/a
	GSPA SLR (EnzBERT Single-Pred)	6/6 (100%)	n/a	n/a	0.4750/0.7058/0.5669	n/a (Stoich. Probe)	n/a
`Price-149` Stratum A ($n = 91$)	CLEAN Base → + SLR Cognate → + Single-Pred → **+ Multi-Pred**	Core Biomass	n/a	n/a	$F_{\max}$: 0.6336 → 0.6405 → 0.6274 → **0.7019**	Macro: 0.4740 → **0.5288**	46 → 46 → 47 → **52** (57.1%)
`Price-149` Stratum B ($n = 55$)	CLEAN Base → + SLR Cognate → + Single-Pred → **+ Multi-Pred**	Catabolic	n/a	n/a	$F_{\max}$: 0.3580 → 0.3753 → 0.3461 → **0.4546**	Macro: 0.3103 → **0.3436**	13 → 13 → **15** (27.3%) → 13 (23.6%)
`Price-149` Stratum C ($n = 2$)	CLEAN Base → + SLR Cognate → + Single-Pred → **+ Multi-Pred**	Secondary	n/a	n/a	$F_{\max}$: 0.5099 → 0.5099 → 0.5175 → **0.5833**	Macro: 0.5000 → 0.5000	1 → 1 → 1 → 1 (50.0%)

Table S6. Summary of 5-fold genus-blocked cross-validation in `BacDive` across carbon-source growth ($n = 2{,}332$ assays, 529 strains) and diagnostic enzymes ($n = 17{,}496$ assays, 3,017 type strains; source: `results/pillar3_bacdive_per_fold_cv.tsv`).

Assay Panel	Classifier	Fold 0 F_1	Fold 1 F_1	Fold 2 F_1	Fold 3 F_1	Fold 4 F_1	5-Fold Mean ± s.d.	Pooled F_1 (BAcc / ROC-AUC)
Carbon Growth ($n = 2{,}332$)	BacDive-AI RF Baseline	0.8709	0.8720	0.8397	0.7174	0.8646	0.8329 ± 0.0659	0.8448 (0.8530/0.9222)
	DiASPora GBDT	0.8582	0.8819	0.8582	0.7817	0.8238	0.8408 ± 0.0390	0.8483 (0.8564/0.9268)
	Traitar L1-SVM	0.8480	0.8765	0.8727	0.7708	0.8653	0.8467 ± 0.0438	0.8571 (0.8638/0.9185)
	GSPA + DiASPora Hybrid	0.8830	0.9048	0.8824	0.7778	0.8602	0.8616 ± 0.0495	0.8710 (0.8783/0.9567)
Diagnostic Enzymes ($n = 17{,}496$)	BacDive-AI RF Baseline	0.8477	0.8371	0.8490	0.8553	0.8387	0.8456 ± 0.0076	0.8458 (0.7911/0.8593)
	DiASPora GBDT	0.8465	0.8579	0.8560	0.8655	0.8493	0.8550 ± 0.0075	0.8550 (0.8108/0.8699)
	Traitar L1-SVM	0.8620	0.8561	0.8562	0.8714	0.8517	0.8595 ± 0.0076	0.8593 (0.8014/0.8715)
	GSPA + DiASPora Hybrid	0.8671	0.8663	0.8718	0.8854	0.8653	0.8712 ± 0.0083	0.8713 (0.8257/0.9097)

S5 Stage 3 neuro-symbolic agent tool schemas, multi-seed stability, and extremophile case study

Stage 3 provides five deterministic, read-only genomic and ontological inspection tools to the open-weight instruction model `Qwen/Qwen3.8-27B-FP8` (`benchmark/agent/run_gspa_agent.py`):

1. `inspect_operon_neighborhood(protein_id, window=5)`: Returns the de-identified genomic neighborhood (±5 genes on the same contig), intergenic distances, strand orientation, and neighbor GO/EC annotations.

2. `inspect_pathway_or_complex_gap(anomaly_id)`: Returns the target pathway or heteromeric complex identifier, fraction of partner steps/subunits present in the genome at $P \geq 0.50$, and list of missing steps/subunits.

3. `query_protein_evidence(protein_id)`: Returns Stage 0 per-channel calibrated probabilities, Pfam/InterPro structural domain annotations, and precomputed RLGC kernel density Bayes factors (BF_{operon}).

4. `check_symbolic_constraints(go_or_ec_term, taxon_id)`: Queries the deterministic ontology bridge (26,911 direct GO taxon rules and 2,715 GO–Uberon/PO/CL anatomical rules linked to 8,331 taxon constraints [4,767 `only_in_taxon`, 3,564 `never_in_taxon`] across 7,722 anatomical and 1,818 taxonomic hierarchy edges) and returns whether the term is valid in `taxon_id`, along with its nearest valid non-root GO ancestors if forbidden.

5. `propose_resolution(action, target_protein_id, resolved_term, confidence, justification)`: Submits the final structured decision (`PROMOTE`, `RELAX`, `DEMOTE`, or `ABSTAIN`), which is deterministically verified by `SymbolicGuardrail` before acceptance.

Supplementary Table S7 reports the per-seed and 3-seed summary of Config 3 (`Qwen/Qwen3.8-27B-FP8 + SymbolicGuardrail`) across seeds 42, 43, and 44 on the 120-anomaly and 50-unique-protein LRO benchmarks (source: `results/pillar4_multiseed_summary.tsv`), and Supplementary Table S8 summarizes the five desert extremophile isolates from the Empty Quarter (*Rub' al Khali*; source: `results/pillar6_empty_quarter_vignette.tsv`).

Table S7. Three-seed evaluation of Stage 3 Config 3 (`Qwen/Qwen3.8-27B-FP8 + SymbolicGuardrail`; source: `results/pillar4_multiseed_summary.tsv`).

Evaluation Run	Random Seed	Precision	Recall	F_1 Score	Accuracy	Violations Accepted	LRO-50 Hit@1 (MRR)
Stage 3 Config 3 (120 Anomalies + LRO-50)	Seed 42	0.9787	0.8519	0.9109	0.9250	0/120 (0.0%)	82.00% (0.8740)
Stage 3 Config 3 (120 Anomalies + LRO-50)	Seed 43	0.9592	0.8704	0.9126	0.9333	0/120 (0.0%)	84.00% (0.8890)
Stage 3 Config 3 (120 Anomalies + LRO-50)	Seed 44	0.9787	0.8519	0.9109	0.9250	0/120 (0.0%)	82.00% (0.8747)
Stage 3 Config 3 (3-Seed Mean ± s.d.)	**Seeds 42, 43, 44**	0.9722 ± 0.0092	0.8581 ± 0.0087	0.9115 ± 0.0008	0.9278 ± 0.0039	$0.00 \pm 0.00\%$	$82.67 \pm 0.94\%$ (0.8792 ± 0.0069)

Table S8. Empty Quarter (*Rub' al Khali*) desert extremophile isolate annotation summary across five bacterial genomes (source: `results/pillar6_empty_quarter_vignette.tsv`).

Empty Quarter Isolate	Taxonomic Clade	Proteins (N)	Raw C_{ess}	Stage 0 C_{ess}	GSPA C_{ess}	Path. Coh. (Raw → GSPA)	pFBA Repairs / Stage 3 Prom.
`EQ_01_Salinicola_sp`	Gammaproteobacteria	3,842	0.8810	0.9120	0.9680	0.4820 → 0.6140	6 repairs / 5 promotions
`EQ_02_Halomonas_sp`	Gammaproteobacteria	4,118	0.8920	0.9210	0.9740	0.4950 → 0.6280	5 repairs / 6 promotions
`EQ_03_Pseudomonas_sp`	Gammaproteobacteria	5,490	0.9140	0.9380	0.9820	0.5190 → 0.6410	4 repairs / 4 promotions
`EQ_04_Bacillus_endophyticus_EQ`	Bacillota	4,625	0.9010	0.9280	0.9760	0.5040 → 0.6320	5 repairs / 7 promotions
`EQ_05_Streptomyces_arid_EQ`	Actinomycetota	7,184	0.8540	0.8960	0.9590	0.4460 → 0.5980	8 repairs / 9 promotions
5-Isolate Mean	**3 Phyla / Classes**	5,051.8	0.8884	0.9190	0.9718	0.4892 → 0.6226	5.6 **repairs** / 6.2 **promotions**